AI Translator Guide

Converting raw country livestock data into the input template

CIAT / CGIAR Alliance, ClimateActionNetZero

Contents

1 What the AI translator is for **2**

2 What the translator will ask, and what it works out for itself **2**

3 One inventory per conversation **3**

4 Large files and fine geography **3**
4.1 The upload limit . . . 3
4.2 The conversation limit . . . 3
4.3 The group limit . . . 4
4.4 Where the aggregation happens . . . 4

5 The three-step workflow **4**
5.1 Step 1: Explore . . . 4
5.2 Step 2: Clarify . . . 4
5.3 Step 3: Emit . . . 5

6 Getting access **6**
6.1 Initial request (first sign-in) . . . 6
6.2 Approval (administrator side) . . . 6
6.3 Subsequent sign-in (after approval) . . . 6
6.4 Staying signed in . . . 6

7 When to use the translator and when to skip it **7**

8 Quality control and limitations **7**

9 Privacy and operational cost **7**
9.1 Where your data goes . . . 7
9.2 Cost . . . 8
9.3 Whether to use it at all . . . 8
9.4 Related material . . . 8

10 What the AI translator is for **8**

11 What the translator will ask, and what it works out for itself **8**

12 One inventory per conversation **9**

13 Large files and fine geography **9**
13.1 The upload limit . . . 9

13.2 The conversation limit . . . 9
13.3 The group limit . . . 10
13.4 Where the aggregation happens . . . 10

14 The three-step workflow 10
14.1 Step 1: Explore . . . 10
14.2 Step 2: Clarify . . . 14
14.3 Step 3: Emit . . . 16

15 Getting access 20
15.1 Initial request (first sign-in) . . . 20
15.2 Approval (administrator side) . . . 22
15.3 Subsequent sign-in (after approval) . . . 22
15.4 Staying signed in . . . 24

16 When to use the translator and when to skip it 24

17 Quality control and limitations 24

18 Privacy and operational cost 27
18.1 Where your data goes . . . 27
18.2 Cost . . . 27
18.3 Whether to use it at all . . . 27
18.4 Related material . . . 27

This page describes what the in-app AI translator does, how to use it for a single inventory, and how access is granted to non-CGIAR email addresses. It complements the user guide and the methodology document with a step-by-step walk-through of the translator workflow.

1 What the AI translator is for

The Cattle Uncertainty App expects its data in a specific Excel template: three sheets (Inventory_Metadata, Parameters, Manure_Management), with controlled vocabularies for sub-categories and manure-management systems, and a fixed column order. Real national inventory data rarely arrives in that shape: column names differ, units may be in litres instead of kilograms, sub-category labels reflect local practice ("Local breed adult cows", "draught oxen"), and uncertainty bounds are sometimes reported as a mean ± standard deviation rather than as an explicit 95% confidence interval.

The AI translator converts raw inventory data, in whatever shape it arrives, into the exact template the tool needs. The user uploads an Excel or CSV file (multi-sheet is supported), answers a small number of clarification questions in natural language, and downloads a filled template ready to upload to the Data Input tab.

The translator does column mapping and unit conversion. It does not do the emissions calculation itself; that happens after the filled template is uploaded.

2 What the translator will ask, and what it works out for itself

There is nothing to prepare and no form to fill in. The user uploads the file, with a note about it if there is something the AI should know, and answers the questions as they arrive. The AI's reply appears as a bubble (the full text behind "Show full reply"); a second bubble says that the questions will come one at a time; then each question is its own bubble, answered in the normal input box with Send. The answers stay on the page until the last one, then go back to the AI in one message, so the AI reflects on all of them at once. "skip" leaves a question to the IPCC default. If the reflection raises new questions, the same thing happens again.

Every answer becomes a recorded decision, and a short line in the conversation says what your answers settled ("Recorded: Inventory year: 2022"). A wrong line is corrected by replying. What the AI read from the file or assumed is listed in the checks workbook. The decisions are sent to the AI with every emission call as binding, they override the file and the defaults, and the finished rows are checked against them: a production system that contradicts one is re-emitted once with the decision quoted, and if it still contradicts, the decision is listed as not applied above the download.

When **Produce template now** is pressed, what can be assumed is assumed and said in one line (the species as mapped from the file, the IPCC edition as in the file, all production systems kept, manure shares from the file where present, the file's ranges where given), and only what cannot be assumed is asked: the inventory year if nothing has settled it, and any column where the header says percent while the values look like fractions. Calves are never asked about: one source category becomes one template category, so a single calves row stays one group. After the last answer the pop-up asks where the files should go, and the template starts from there.

The country name and the inventory year are also required, and are quick to state if they are not in the file.

3 One inventory per conversation

Each finished template describes one country and one inventory year, because the Inventory_Metadata sheet carries a single country field. Two inventories therefore need two templates, two conversations and two runs of the tool.

Start a new conversation for the second inventory rather than continuing the first. A conversation carries the sub-category vocabulary, unit conventions and manure system mapping agreed for the inventory already in it, and those are exactly the things that can be carried across to the next one without anyone noticing: the resulting template is still well formed, and the error is only visible by comparing it against the source file.

Two people working on different inventories at the same time should also expect the panel to feel slower than it does for one person alone, because the app serves everyone from a single process. The translation itself is unaffected, but replies queue behind each other. Where several inventories are being prepared in one session, it is worth staggering the work rather than starting every translation at once.

4 Large files and fine geography

Two separate limits apply, and they catch different files. One is about the size of the file you upload, the other about how many groups the inventory describes. A file can pass one and fail the other.

4.1 The upload limit

The panel accepts `.xlsx`, `.xlsm`, `.xls` and `.csv`, up to **8 MB**. A larger file is refused by the browser before it reaches the tool, with a message saying so.

National calculator workbooks are often well over this, and usually for reasons that have nothing to do with the inventory: feed composition tables, municipality lookups, macros and embedded charts. The fix is to copy the sheets that actually hold the cattle data into a new workbook and upload that. It is normal for a 20 MB national calculator to become a 200 KB file with two sheets in it.

4.2 The conversation limit

Even inside 8 MB, a file can hold more data than the conversation can carry. The panel shows an estimated token count for each sheet before anything is sent, and warns if the total is too large. Untick the sheets the translator does not need: reference tables, vocabulary lists, calculation sheets whose results already appear elsewhere.

Nothing is charged until you press send, so a warning at this stage costs nothing to act on.

4.3 The group limit

The tool simulates about **400 groups**, where a group is one combination of `cattle_type`, `aggregation_level` and `sub_category`. An inventory with 1,121 municipalities and 9 animal sub-categories is roughly 10,000 groups, which no amount of translating will make runnable.

Decide the geographic level before you translate, not after. Translating at municipality level and aggregating afterwards wastes the translation, because the aggregation has to happen in the source data and the file then has to be translated again.

The translator is told about this limit and should raise the question itself, as the first item in its list of ambiguities, when your file carries more than one geographic level. If it does not raise it, raise it yourself: tell it which level to use, in the same message where you answer its other questions.

4.4 Where the aggregation happens

In your own data, before you upload. Most national calculators already contain the coarser level: a department total row alongside the municipality rows. If yours does, point the translator at those rows and ignore the rest.

If it does not, aggregate before uploading: add the populations together, and take population-weighted averages of every other parameter. The user guide describes this under "How many groups the tool can run", along with the reason that splitting a large inventory into several runs and combining the results afterwards gives the wrong confidence interval.

5 The three-step workflow

The translator works in three explicit passes: Explore, Clarify, Emit, rather than attempting all three at once. This separation prevents the AI from silently dropping rows or substituting catalogue defaults for values the user supplied.

5.1 Step 1: Explore

After the file is read, the panel lists its sheets with their size and lets the user untick reference or lookup sheets the AI does not need before sending; fewer sheets mean a faster first reply and a cheaper run. The AI's first response is then a structured exploration report with four labelled sections:

- **A. File shape.** For each sheet in the upload, the AI classifies the layout pattern (column-oriented, wide-stacked, parameter-labelled, reference-table, or calc-sheet).
- **B. Inventory of values found.** A table listing every (parameter, sub-category) pair the AI could identify in the file, with the source sheet, row, column, central value, lower and upper bounds where present, units, and any qualifier (such as "Local breed only" or "Cross breed weighted average").
- **C. Inventory of gaps.** IPCC catalogue parameters that the file does not provide. These will need to be filled with IPCC defaults at emission time.
- **D. Ambiguities to resolve.** Questions the user must answer before the template can be produced: sub-category vocabulary mapping, unit conversions, biological zeros, manure-management code interpretation, breed disaggregation, and similar.

The exploration report is the persistent ground-truth artefact that drives the rest of the workflow. The user is expected to read it carefully: section B in particular should list every parameter present in the source file. A parameter missing from section B should be raised in the chat before proceeding.

5.2 Step 2: Clarify

The user answers the questions in section D in plain natural-language chat messages. There is no required template for the answers; the AI parses them and updates its internal mapping. Typical clarifications include:

- Sub-category vocabulary mapping (e.g., "Cows" → `other_cows`; "Adult males" → `bulls`). One source category becomes one template category: a single calves row in the file becomes the pooled `calves` code, never a male and female split the file does not make.
- Unit conversions (litres of milk to kilograms using the cow-milk density 1.03 kg/L; digestibility expressed as a fraction 0.55 versus a percentage 55%).
- Biological zeros (Milk = 0 for non-lactating sub-categories; hours of work = 0 for non-oxen).
- Manure-management code interpretation (e.g., a file-level "PIT" entry → `liquid_slurry`; "DL" → `dry_lot`).
- Whether the file's mean ± SD reporting should be converted to mean ± 1.96 · SD for the 95% confidence interval.

If the user wants to add data that is not in the file (typically national population numbers), the values can be typed directly into the chat. They will be incorporated into the final template alongside the file-derived values.

Revising a produced template. After a template has been produced, the link **Back to the conversation** keeps the answers and the history, and a change can be typed as a reply. In that revision, and only there, a button **Upload a past translation** sits above the reply box: a template produced by the tool, corrected in Excel and uploaded there is read next to the source file, which stays in the conversation. The checks are run on it at once, a note box carries the user's comments into the chat, and the AI is told that this is a past translation whose values outrank the original file and the earlier answers. Pressing **Produce template now** then builds the template from the corrected values.

5.3 Step 3: Emit

When the section D questions are answered, the user clicks **Produce template now**. The AI translates section B (file-derived values), section C (catalogue gaps to be filled with IPCC defaults), and the biological zeros confirmed in section D, mechanically, into the JSON template, one production system at a time.

The run is a job the page does not have to stay open for. A progress bar shows the system being produced, the time elapsed and an estimate of the time left (from the size of each system and the speed of earlier runs). When the files are ready they are delivered by themselves: with **Choose where to save** pressed beforehand (Chrome and Edge), all three files are written into that folder; otherwise the data workbook downloads by itself and the other two files get a button. If the page was closed before the end, the three files are emailed to the signed-in address, and they also stay in the app until the server restarts.

Three files come out of a translation, named after the country, the inventory year and the time: the **data workbook** (the template, with a filter on every column, flagged cells coloured red, orange or blue, and a Check and a Why column at the right of each sheet), the **checks workbook** (every point to review, failures first, with a column to record what was decided, plus the decisions and a summary), and the **conversation document** (the questions and the answers, raw).

Above the buttons sits a **quality preview**: the same checks the QA/QC tab will run after upload (pass, warn and fail counts with the failing rows) plus every consistency note that compares rows with each other and with the source file. The notes name the places where a translation typically goes wrong: a source category missing from the template (every category of the file must be a group unless a merge was agreed), a value in the Parameters sheet that differs from the time series for the inventory year, a percentage written as a fraction or the reverse, a population counted under two groups, a value tagged as read from the file that the file does not contain, manure shares that do not sum to 100 %, one manure system given different coefficients in different production systems, and a range or a distribution the file did not give and the IPCC catalogue supplied. Each note is a sentence to check against the source file. The user can go back to the conversation and correct, or download and deal with the points in the checks workbook and on the QA/QC tab.

The downloaded file follows the official template format exactly. After download, the user should open the file and skim the `data_source` column of the Parameters sheet to confirm that values from the source file are tagged `user_file`, IPCC catalogue values are tagged `ipcc_default`, and explicit zero values are tagged

`biological_zero` or `placeholder`. This is the recommended spot-check for catching translation failures before the simulation runs.

6 Getting access

The translator is gated behind a magic-link sign-in flow. Every address requires one-time manual approval by the administrator, including CGIAR addresses. Each translation is billed to the project, so access is granted by name rather than by domain. Approval is needed once; after that the browser stays signed in. The full sequence is as follows.

6.1 Initial request (first sign-in)

1. Open the **Resources tab** of the Cattle Uncertainty App and locate the AI Translator card (the first item, identified by the green left-edge accent).
2. Enter the email address into the sign-in field and click **Send sign-in link**.
3. Within 1–2 minutes, an email arrives from `noreply@cattle-uncertainty.app` with subject "Sign in to the IPCC Cattle GHG Uncertainty tool". The link in it stays valid for 30 days and may be clicked more than once; after the first successful click the browser stays signed in, so the link is only needed once per browser.
4. Click the link. The app opens in a browser tab and recognises that the email is not yet approved. A confirmation message is displayed:

   > Thanks: your request to access the AI translator (your.email@example.org) has been forwarded to the administrator for approval. You'll receive an email once approved.

   At the same moment, the administrator receives a notification email summarising the request (email address, timestamp).

6.2 Approval (administrator side)

5. The administrator reviews the request and, if appropriate, adds the email address to the approved-users list and redeploys the app. The administrator typically responds the same working day.
6. Once added, the user automatically receives a welcome email confirming approval. The email contains a direct link back to the app and a brief reminder of the workflow.

6.3 Subsequent sign-in (after approval)

7. Return to the AI Translator card on the Resources tab.
8. Enter the same email address again and click **Send sign-in link**. A fresh magic-link email arrives within 1–2 minutes.
9. Click the new link. The app verifies the email is now on the approved list, drops a long-lived sign-in cookie in the browser, and unlocks the chat panel.
10. On subsequent visits from the same browser, the cookie keeps the user signed in. No additional magic-link step is required unless the user clears site data or switches browsers.

6.4 Staying signed in

The app remembers you in the browser you signed in with. A different browser or machine, or clearing your site data, means asking for a new sign-in link. A private or incognito window forgets everything when you close it, so it needs a new link every time. Use a normal window, especially if sign-in emails are slow to reach you.

Your conversation is safe either way. It is saved against your email address, not in the browser, so it comes back when you sign in again.

7 When to use the translator and when to skip it

The AI translator is intended for the common case in which raw inventory data does not match the tool's template. It is most useful when:

- The user's data is in a non-standard shape (multi-sheet workbook with section headers, form-style layout, free-text comments interspersed with values).
- Column names differ from the template (e.g., `cattle_pop` instead of `N`, `live_wt_lb` instead of `BW`).
- Units are inconsistent (mix of kg and lb, fractions and percentages, litres and kilograms of milk).
- IPCC defaults are needed for parameters the user does not measure.

The translator is not required when the user already has a filled template: in that case, upload directly via the Data Input tab. The translator is also not the appropriate tool for converting an emissions-calculation result; it works on input parameters only.

8 Quality control and limitations

The translator performs column mapping and unit conversion. It does not validate the underlying values against external reference data, and it cannot detect systematic errors in the source file (e.g., a transcription mistake in the original survey). Two checks are recommended after every translation:

- **Read the quality preview before downloading.** It lists the checks that will fail after upload and the consistency notes described under Step 3. On the first real translation (a 27-group national inventory) every error the AI had introduced appeared in that list.
- **Spot-check the key numbers.** Compare population, body weight, and milk yield in the downloaded template against the corresponding cells in the source file. Any discrepancy indicates a translation error to be raised in the chat.
- **Review the QA/QC tab after upload.** Any value the translator filled from an IPCC default is flagged with `data_source = "ipcc_default"` in the QA tab, allowing the user to review whether the catalogue default is appropriate for the country context. For body weight (`BW`), the QA tab also runs a continental-default deviation check (Annex Tables 10A.1 / 10A.2 / 10A.3) and warns on deviations larger than ±50%.
- **The workbook records its provenance.** The Notes cell of `Inventory_Metadata` carries the app version, a fingerprint of the instructions the AI worked from, a fingerprint of the IPCC defaults table, the model, and the date, so a file that comes back for support can be traced to the defaults it was built with.
- **Resumed conversations are checked against the current defaults.** If the IPCC defaults or the template changed since a saved conversation was held, the panel says so and asks for the file to be uploaded again before a template can be produced; the earlier discussion would otherwise carry the old numbers into the workbook.

9 Privacy and operational cost

9.1 Where your data goes

When a file is uploaded, its contents are sent to the AI service over an encrypted connection. That is how the translation is done, and it is the main thing to be aware of before uploading anything.

A copy of the conversation, including the uploaded data and the user's email address, is then written to a file on the app server. It is kept so that a user can close the tab and resume where they left off. It is removed in two cases: when the user clicks **Reset conversation**, or when the server restarts after a period without use. There is no scheduled deletion beyond that, so **Reset conversation is the reliable way to remove it**.

Separately, a usage log records the date, the user's email address, the model and the token count of each call. It holds no file contents and exists so the administrator can monitor spending.

The AI service applies its own terms to the data it receives. Those terms are the service provider's and should be consulted directly rather than summarised here.

9.2 Cost

The translator is hosted by the GMH project on a dedicated AI service account, and the user is not billed for usage. A monthly global spending cap is in place; if the cap is reached, an explicit "monthly cap reached, contact administrator" message is shown and translation requests pause until the next cycle.

9.3 Whether to use it at all

The IPCC inventory data the translator handles is generally non-sensitive (it is destined for UNFCCC reporting, which is a public-record process). For ministries with internal policies restricting upload of pre-publication data to external AI services, the translator should not be used; the template can instead be filled manually using the IPCC defaults provided in the user guide.

9.4 Related material

- User Guide, §"AI-assisted data preparation": the same workflow described in the printable user guide, including the non-CGIAR step-by-step.
- Methodology document: full equations, parameter definitions, and references for everything the translator outputs.
- Find out more: Correlations, the companion topic page covering the correlation modes on the Simulate tab.

Generated by scripts/build_help_docs.R from documentation/source/ai_translator.Rmd.

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This page explains what the in-app AI translator does, how to use it for one inventory, and how access is given. It complements the user guide and the methodology document with a walk-through of the translator, step by step, with a picture of each screen.

10 What the AI translator is for

The tool needs its data in one Excel template. The template has three sheets (Inventory_Metadata, Parameters and Manure_Management), fixed names for the animal classes and the manure systems, and a fixed order of columns. Real inventory data rarely looks like that. Column names differ. Milk may be in litres instead of kilograms. Animal classes have local names ("Local breed adult cows", "draught oxen"). Uncertainty is sometimes given as a mean plus or minus a standard deviation rather than as a 95% confidence interval.

The AI translator turns your data, in whatever shape it arrives, into the exact template the tool needs. You upload an Excel or CSV file (several sheets are fine), answer a small number of questions in plain language, and download a filled template ready for the Data Input tab.

The translator only maps columns and converts units. It does not calculate emissions. That happens after you upload the filled template.

11 What the translator will ask, and what it works out for itself

There is nothing to prepare and no form to fill in. You upload the file, add a note if there is something the AI should know, and answer the questions as they come.

The AI's first reply appears as a bubble (the full text is behind "Show full reply"). A second bubble says that the questions will come one at a time. Each question is then its own bubble. You answer in the normal

box and press Send. Your answers stay on the page until the last one. Then they go back to the AI in one message, so it can think about all of them together. Typing "skip" leaves a question to the IPCC default. If the AI has new questions after reading your answers, the same thing happens again.

Every answer becomes a recorded decision. A short line in the conversation says what your answers settled, for example "Recorded: Inventory year: 2022". If a line is wrong, reply with the correction. What the AI read from the file or assumed on its own is listed in the checks workbook. Your decisions are sent to the AI with every call that writes the template, as rules that override the file and the defaults. The finished rows are checked against them. A production system that contradicts a decision is written again once, with the decision quoted. If it still contradicts, the decision is listed as "not applied" above the download button.

When you press **Produce template now**, the AI assumes what can be assumed and says so in one line: the species as mapped from the file, the IPCC edition as given in the file, all production systems kept, manure shares from the file where present, the file's ranges where given. It asks only what it cannot assume: the inventory year if nothing has settled it, and any column where the header says percent while the values look like fractions. Calves are never asked about. One category in your file becomes one category in the template, so a single calves row stays one group. After the last answer, a pop-up asks where the files should go, and the template is produced from there.

The country name and the inventory year are also needed. If they are not in the file, say them in the notes box or in a reply.

12 One inventory per conversation

One finished template describes one country and one inventory year, because the Inventory_Metadata sheet has one country field. Two inventories need two templates, two conversations and two runs of the tool.

Start a new conversation for the second inventory. Do not continue the first one. A conversation carries the animal-class names, the units and the manure-system mapping agreed for the inventory already in it. Those are exactly the things that could slip across to the next inventory without anyone noticing: the template would still look fine, and the error would only show when compared with the source file.

Two people translating at the same time will find the panel slower than usual, because the app serves everyone from one process. The translation itself is not affected; the replies queue behind each other. When several inventories are prepared in one session, stagger the work rather than starting every translation at once.

13 Large files and fine geography

Two separate limits apply, and they catch different files. One is about the size of the file you upload. The other is about how many groups the inventory describes. A file can pass one and fail the other.

13.1 The upload limit

The panel accepts `.xlsx`, `.xlsm`, `.xls` and `.csv`, up to **8 MB**. A larger file is refused by the browser before it reaches the tool, with a message saying so.

National calculator workbooks are often well over this, usually for reasons that have nothing to do with the inventory: feed composition tables, municipality lookups, macros and charts. The fix is to copy the sheets that hold the cattle data into a new workbook and upload that. A 20 MB national calculator often becomes a 200 KB file with two sheets.

13.2 The conversation limit

Even inside 8 MB, a file can hold more data than one conversation can carry. After the upload, the panel shows an estimated size for each sheet before anything is sent, and warns if the total is too large. Untick

the sheets the translator does not need: reference tables, vocabulary lists, calculation sheets whose results already appear elsewhere.

Nothing is charged until you press send, so a warning at this stage costs nothing to act on.

13.3 The group limit

The tool simulates about **400 groups**, where a group is one combination of `cattle_type`, `aggregation_level` and `sub_category`. An inventory with 1,121 municipalities and 9 animal classes is roughly 10,000 groups, which no translation will make runnable.

Decide the geographic level before you translate, not after. Translating at municipality level and aggregating afterwards wastes the translation, because the aggregation has to happen in the source data, and the file then has to be translated again.

The translator knows this limit and should raise the question itself, as the first of its questions, when your file carries more than one geographic level. If it does not, raise it yourself: tell it which level to use, in the same message as your other answers.

13.4 Where the aggregation happens

In your own data, before you upload. Most national calculators already contain the coarser level: a department total next to the municipality rows. If yours does, point the translator at those rows and ignore the rest.

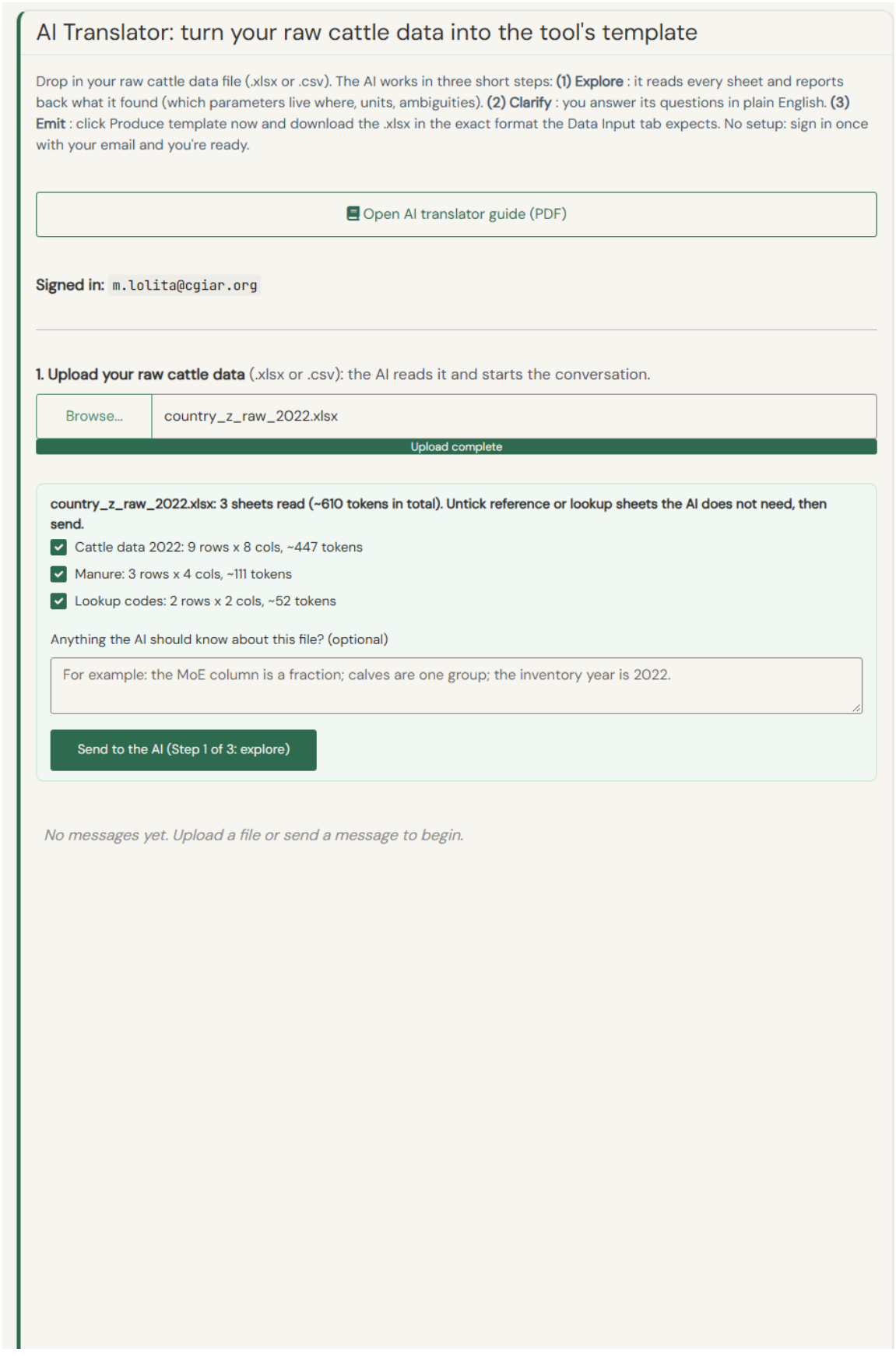
If it does not, aggregate before uploading: add the populations, and take population-weighted averages of every other parameter. The user guide explains this under “How many groups the tool can run”, together with the reason why splitting a large inventory into several runs and combining the results afterwards gives the wrong confidence interval.

14 The three-step workflow

The translator works in three passes: Explore, Clarify, Emit. It does not try to do everything at once. The separation stops the AI from quietly dropping rows or putting catalogue defaults where you supplied values.

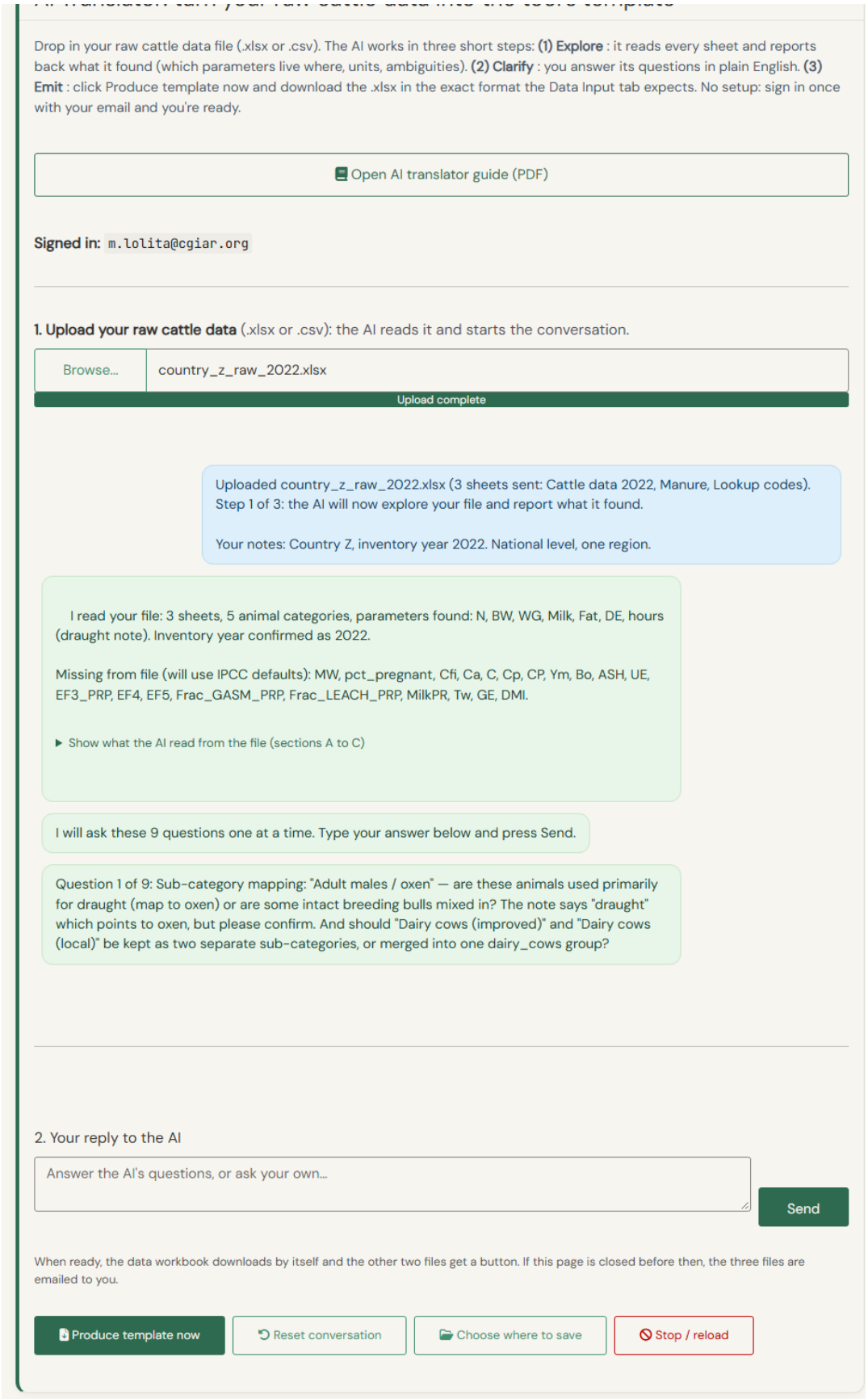
14.1 Step 1: Explore

Click **Browse** in the translator card and pick your file. The panel lists the sheets it found, with the size of each, and lets you untick the reference or lookup sheets the AI does not need. Fewer sheets mean a faster first reply and a cheaper run. There is also a box for notes: anything the AI should know, such as the country, the year or the units. Then click **Send to the AI (Step 1 of 3: explore)**.


AI Translator: turn your raw cattle data into the tool's template
Drop in your raw cattle data file (.xlsx or .csv). The AI works in three short steps: (1) Explore : it reads every sheet and reports back what it found (which parameters live where, units, ambiguities). (2) Clarify : you answer its questions in plain English. (3) Emit : click Produce template now and download the .xlsx in the exact format the Data Input tab expects. No setup: sign in once with your email and you're ready.
Open AI translator guide (PDF)
Signed in: m.lolita@cgiar.org
1. Upload your raw cattle data (.xlsx or .csv): the AI reads it and starts the conversation.
Browse...
country_z_raw_2022.xlsx
Upload complete
country_z_raw_2022.xlsx: 3 sheets read (~610 tokens in total). Untick reference or lookup sheets the AI does not need, then send.
Cattle data 2022: 9 rows x 8 cols, ~447 tokens
Manure: 3 rows x 4 cols, ~111 tokens
Lookup codes: 2 rows x 2 cols, ~52 tokens
Anything the AI should know about this file? (optional)
For example: the MoE column is a fraction; calves are one group; the inventory year is 2022.
Send to the AI (Step 1 of 3: explore)
No messages yet. Upload a file or send a message to begin.

After about a minute, the AI replies with a short summary: the sheets, the animal classes and the parameters it found, and the parameters missing from the file. The full exploration report is behind "Show what the AI read from the file". It has four parts:

- **A. File shape.** For each sheet, how it is laid out (columns, wide blocks, parameter labels, a reference table, a calculation sheet).
- **B. Values found.** Every parameter and animal class the AI could identify, with the sheet, row and column, the central value, the lower and upper bounds where present, the units, and any qualifier (such as "Local breed only").
- **C. Gaps.** The IPCC parameters your file does not provide. These are filled with IPCC defaults when the template is written.
- **D. Questions.** What the AI needs you to settle before the template can be produced: animal-class names, unit conversions, biological zeros, manure-system codes, breed splits, and the like.

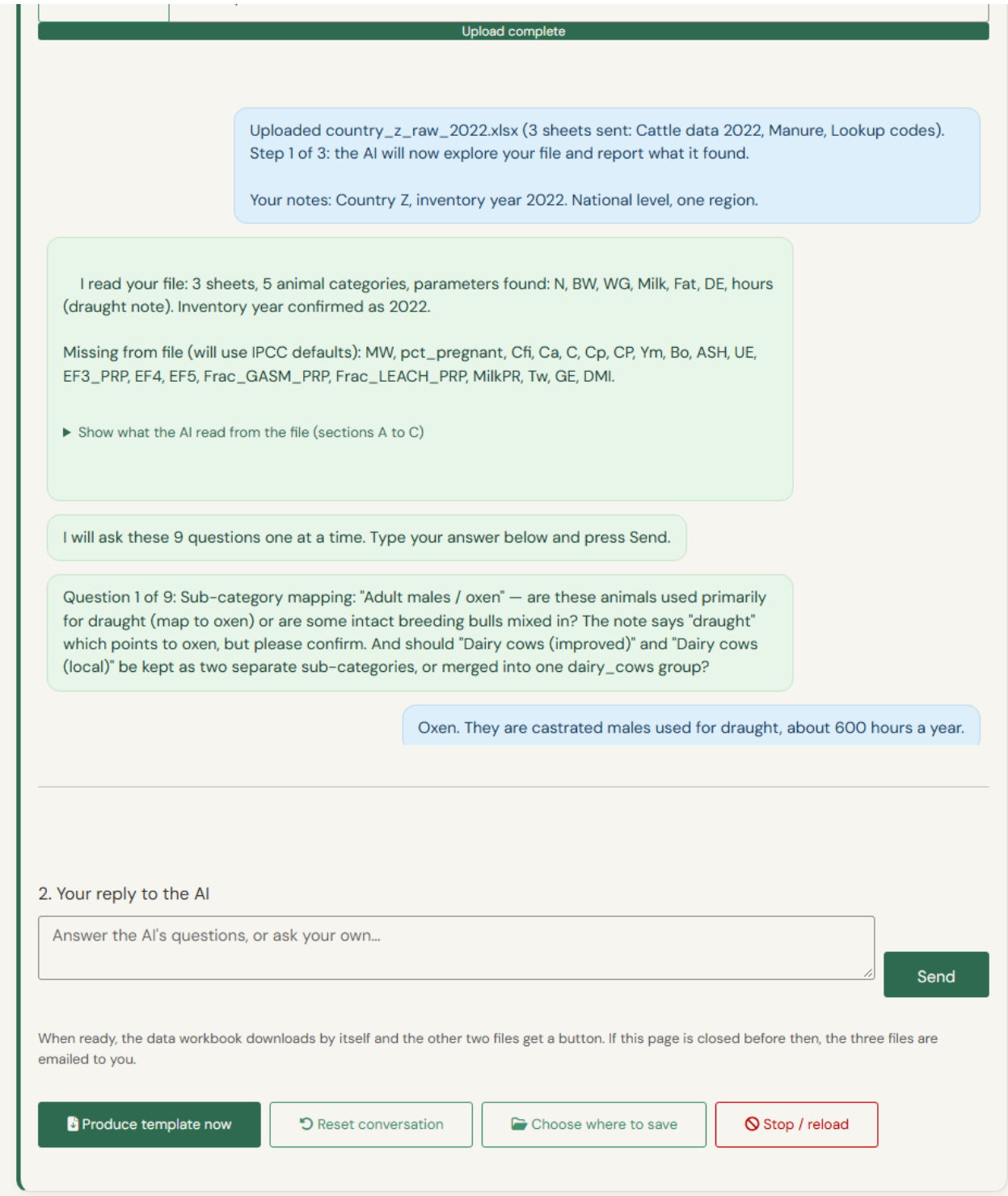

Drop in your raw cattle data file (.xlsx or .csv). The AI works in three short steps: (1) Explore : it reads every sheet and reports back what it found (which parameters live where, units, ambiguities). (2) Clarify : you answer its questions in plain English. (3) Emit : click Produce template now and download the .xlsx in the exact format the Data Input tab expects. No setup: sign in once with your email and you're ready.
Open AI translator guide (PDF)
Signed in: m.lolita@cgiar.org
1. Upload your raw cattle data (.xlsx or .csv): the AI reads it and starts the conversation.
Browse...
country_z_raw_2022.xlsx
Upload complete
Uploaded country_z_raw_2022.xlsx (3 sheets sent: Cattle data 2022, Manure, Lookup codes). Step 1 of 3: the AI will now explore your file and report what it found.
Your notes: Country Z, inventory year 2022. National level, one region.
I read your file: 3 sheets, 5 animal categories, parameters found: N, BW, WG, Milk, Fat, DE, hours (draught note). Inventory year confirmed as 2022.
Missing from file (will use IPCC defaults): MW, pct_pregnant, Cfi, Ca, C, Cp, CP, Ym, Bo, ASH, UE, EF3_PRP, EF4, EF5, Frac_GASM_PRP, Frac_LEACH_PRP, MilkPR, Tw, GE, DMI.
▶ Show what the AI read from the file (sections A to C)
I will ask these 9 questions one at a time. Type your answer below and press Send.
Question 1 of 9: Sub-category mapping: "Adult males / oxen" — are these animals used primarily for draught (map to oxen) or are some intact breeding bulls mixed in? The note says "draught" which points to oxen, but please confirm. And should "Dairy cows (improved)" and "Dairy cows (local)" be kept as two separate sub-categories, or merged into one dairy_cows group?
2. Your reply to the AI
Answer the AI's questions, or ask your own...
Send
When ready, the data workbook downloads by itself and the other two files get a button. If this page is closed before then, the three files are emailed to you.
Produce template now
Reset conversation
Choose where to save
Stop / reload

This report is the reference for the rest of the work. Read it, and part B in particular: it should list every parameter present in your file. If a parameter is missing from part B, say so in the chat before going on.

14.2 Step 2: Clarify

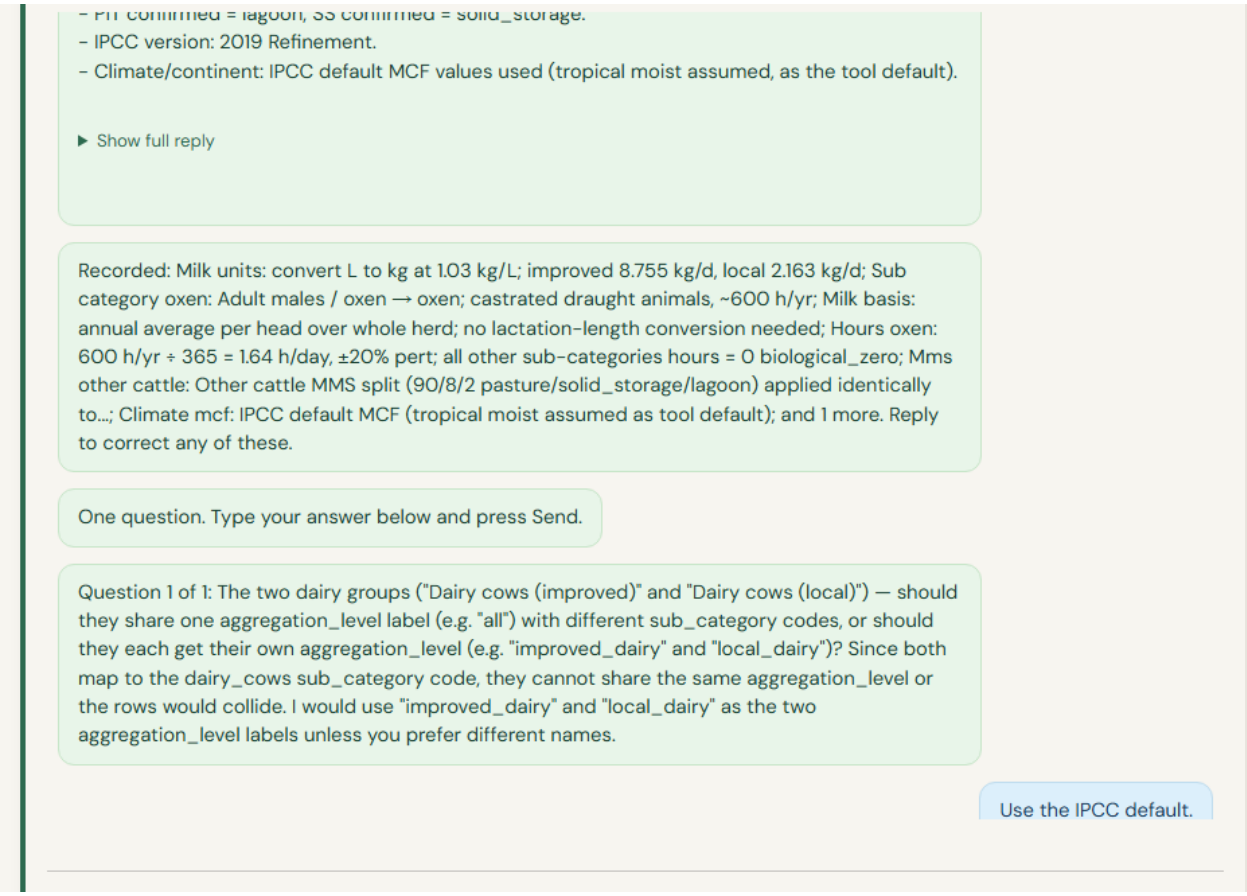
The questions of part D come one at a time. Answer each one in plain language in the reply box and press Send. There is no format to follow; the AI reads your answer and updates its mapping. Typical answers:

- Animal-class names ("Cows" means `other_cows`; "Adult males" means `bulls`). One category in the file becomes one category in the template: a single calves row becomes the pooled `calves` code, never a male and female split the file does not make.
- Unit conversions (litres of milk to kilograms at 1.03 kg per litre; digestibility given as 0.55 or as 55%).
- Biological zeros (Milk = 0 for classes that do not lactate; hours of work = 0 for anything but oxen).
- Manure-system codes ("PIT" means `liquid_slurry`; "DL" means `dry_lot`).
- Whether a mean plus or minus a standard deviation should be turned into a 95% confidence interval (mean plus or minus 1.96 standard deviations).



Data that is not in the file, typically national population numbers, can be typed straight into the chat. It goes into the template next to the values from the file.

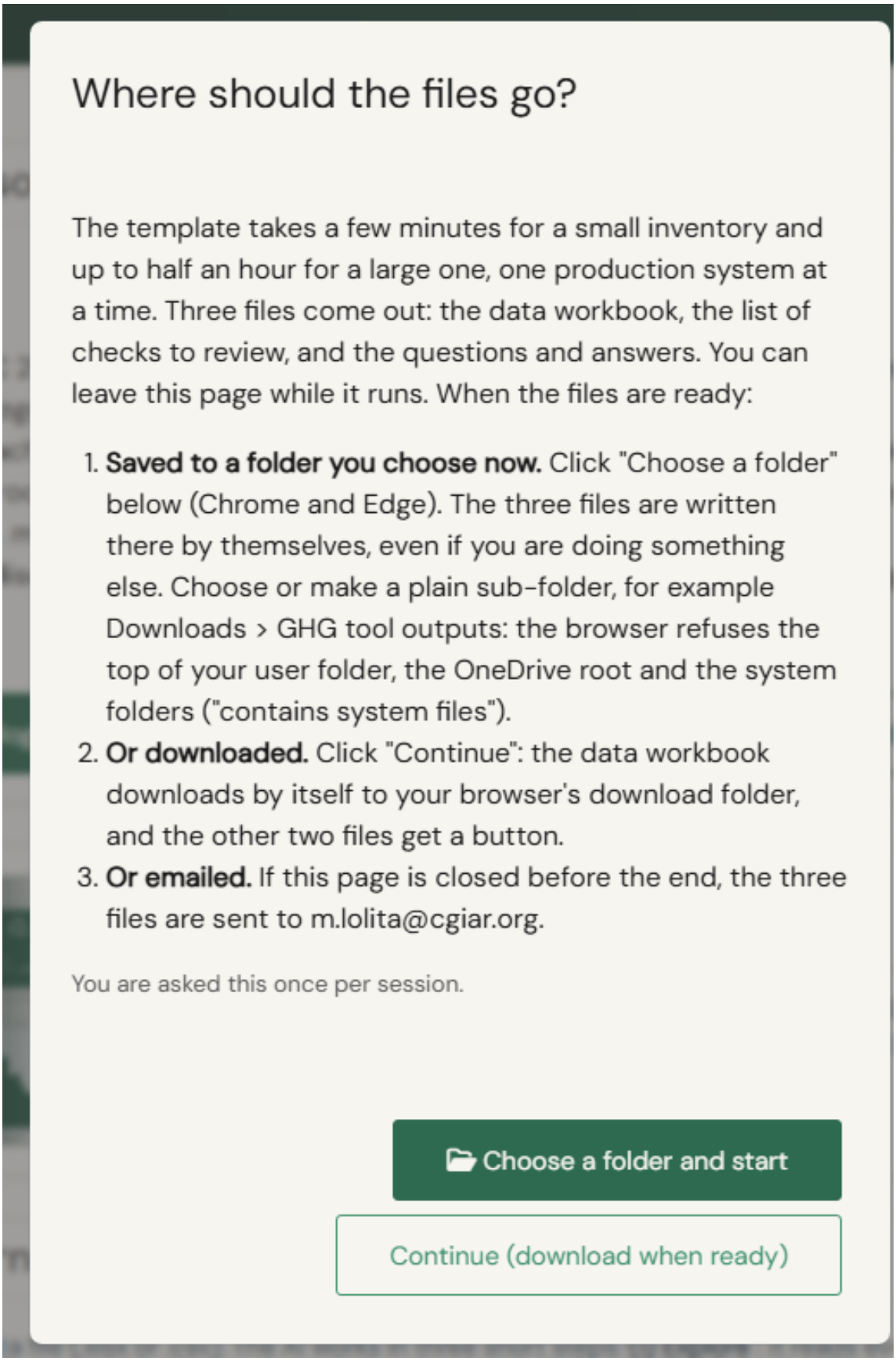
When the last question is answered, the AI replies with a summary and one “Recorded:” line per decision. Check these lines and reply if one is wrong.



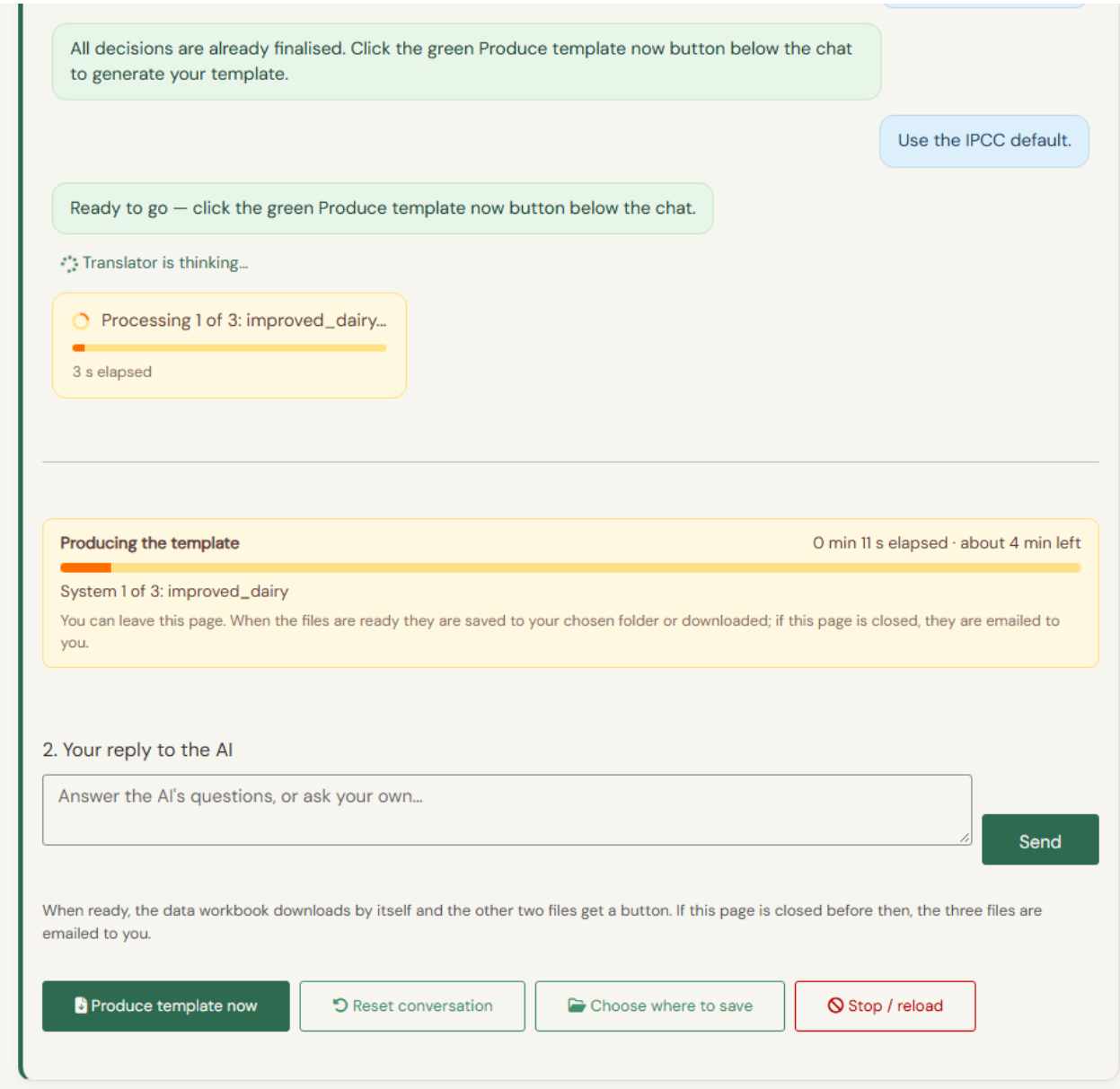
Revising a produced template. After a template has been produced, the link **Back to the conversation** keeps your answers and the history, and a change can be typed as a reply. In that revision, and only there, a button **Upload a past translation** sits above the reply box. A template produced by the tool, corrected in Excel and uploaded there, is read next to the source file, which stays in the conversation. The checks run on it at once, a note box carries your comments into the chat, and the AI is told that this is a past translation whose values outrank the original file and the earlier answers. Pressing **Produce template now** then builds the template from the corrected values.

14.3 Step 3: Emit

When the questions are answered, click **Produce template now**. A pop-up asks where the files should go. You have three options: a folder you choose now (Chrome and Edge; the three files are written there when ready), a download (the data workbook downloads by itself and the other two files get a button), or email (if the page is closed before the end, the three files are sent to the address you signed in with). You are asked once per session.


Where should the files go?
The template takes a few minutes for a small inventory and up to half an hour for a large one, one production system at a time. Three files come out: the data workbook, the list of checks to review, and the questions and answers. You can leave this page while it runs. When the files are ready:
1. Saved to a folder you choose now. Click "Choose a folder" below (Chrome and Edge). The three files are written there by themselves, even if you are doing something else. Choose or make a plain sub-folder, for example Downloads > GHG tool outputs: the browser refuses the top of your user folder, the OneDrive root and the system folders ("contains system files").
2. Or downloaded. Click "Continue": the data workbook downloads by itself to your browser's download folder, and the other two files get a button.
3. Or emailed. If this page is closed before the end, the three files are sent to m.lolita@cgiar.org.
You are asked this once per session.
Choose a folder and start
Continue (download when ready)

The AI then writes the template from part B (values from your file), part C (gaps filled with IPCC defaults) and the biological zeros settled in part D, one production system at a time. The run is a job the page does not have to stay open for. A progress bar shows the system being produced, the time elapsed and an estimate of the time left, from the size of each system and the speed of earlier runs. A small inventory takes a few minutes; a large one up to half an hour.



When the files are ready, the card shows a quality preview and three download buttons. The three files are named after the country, the inventory year and the time:

- the **data workbook**: the template, with a filter on every column, flagged cells coloured red, orange or blue, and a Check and a Why column at the right of each sheet;
- the **checks workbook**: every point to review, failures first, with a column to record what you decided, plus the decisions and a summary;
- the **conversation document**: the questions and the answers, as they were.

Noted — no change needed, all decisions are already locked in using IPCC defaults where applicable. Everything is ready.

Click the green Produce template now button below the chat to start emission.

Recorded: Production systems kept: improved_dairy, local_dairy, non_dairy (IPCC default naming). Reply to correct any of these.

Use the IPCC default.

All decisions are already finalised. Click the green Produce template now button below the chat to generate your template.

Use the IPCC default.

Ready to go — click the green Produce template now button below the chat.

Template ready. Emitted 135 parameter rows, 15 manure_management rows, and 5 time-series rows across 3 aggregation_levels (improved_dairy, local_dairy, non_dairy). Click the green Download button below to save the .xlsx, then upload it on the 1. Data Input tab.

Your template is ready. Click the green button to save it, then open it and check the AI's work before uploading it on the 1. Data Input tab.

Quality preview of this template. 135 parameter rows, 16 manure rows.

497 pass | 1 warn | 0 fail | 34 consistency notes

Consistency notes to check against your file. These compare rows with each other. Each one is a place where the AI may have copied or invented a value.

- improved_dairy / dairy_cows / Milk: Milk = 8.755 is marked as taken from your file but no matching number is in it (2 such value(s) in all). Either the value was altered, or it was computed and should be marked user_chat rather than user_file.
- local_dairy / dairy_cows / Milk: Milk = 2.163 is marked as taken from your file but no matching number is in it (2 such value(s) in all). Either the value was altered, or it was computed and should be marked user_chat rather than user_file.
- category coverage: Source categories: 5; template groups: 5; every source category is in the template.
- Parameters vs time series: Parameters and Parameter_TimeSeries agree for the inventory year 2022 on every group and parameter checked.
- improved_dairy / dairy_cows / MCF_pct: MCF_pct came without a range; plus or minus 25% (pert) was applied as a project assumption. Replace it if your inventory has its own.
- improved_dairy / dairy_cows / EF3: EF3 came without a range; a factor of two either way (lognormal), the spread of the IPCC 2006 Table 10.21 ranges, was applied. Replace it if your inventory has its own.

Download template (.xlsx)

Download the checks (Excel) | Download the conversation (Word)

Three files: the data workbook (flagged cells are coloured, with a Check and a Why column at the right of each sheet, and filters on every column), the list of checks to review, and the questions and answers.

Back to the conversation (your answers are kept) Start a new translation

The **quality preview** above the buttons runs the same checks the QA/QC tab will run after upload (pass, warn and fail counts, with the failing rows) plus the consistency notes that compare rows with each other and with the source file. The notes name the places where a translation typically goes wrong: a category in the file that is not in the template (every category must be a group unless a merge was agreed), a value in the Parameters sheet that differs from the time series for the inventory year, a percentage written as a fraction or the reverse, a population counted under two groups, a value marked as read from the file that the file does not contain, manure shares that do not add up to 100%, one manure system given different coefficients in different production systems, and a range or a distribution the file did not give and the IPCC catalogue supplied. Each note is one thing to check against the source file. You can go back to the conversation and correct it, or download the files and deal with it in the checks workbook and on the QA/QC tab.

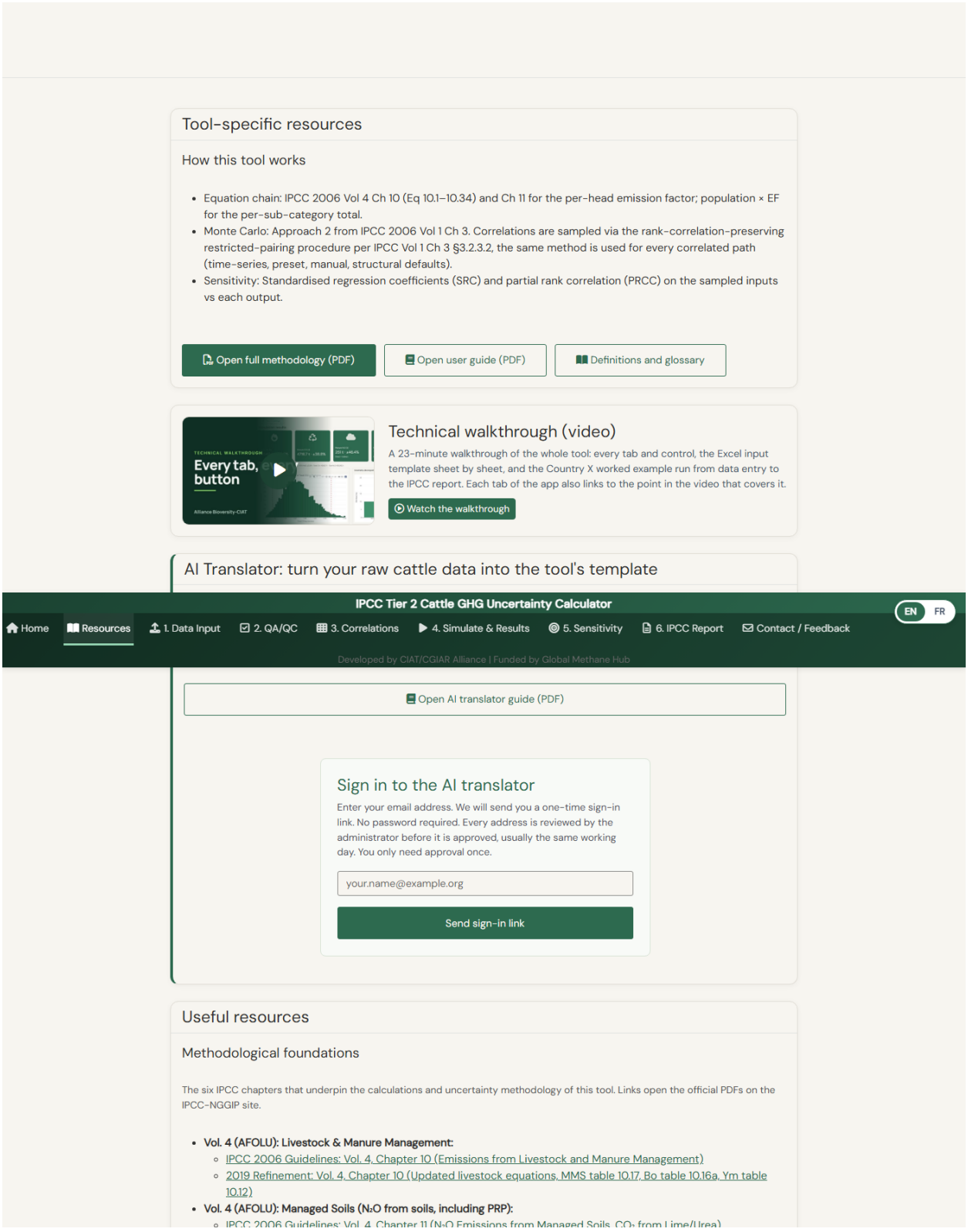
The data workbook follows the official template exactly. After downloading, open it and look at the `data_source` column of the Parameters sheet: values from your file are marked `user_file`, IPCC defaults are marked `ipcc_default`, and explicit zeros are marked `biological_zero` or `placeholder`. This is the quickest way to catch a translation error before the simulation runs.

15 Getting access

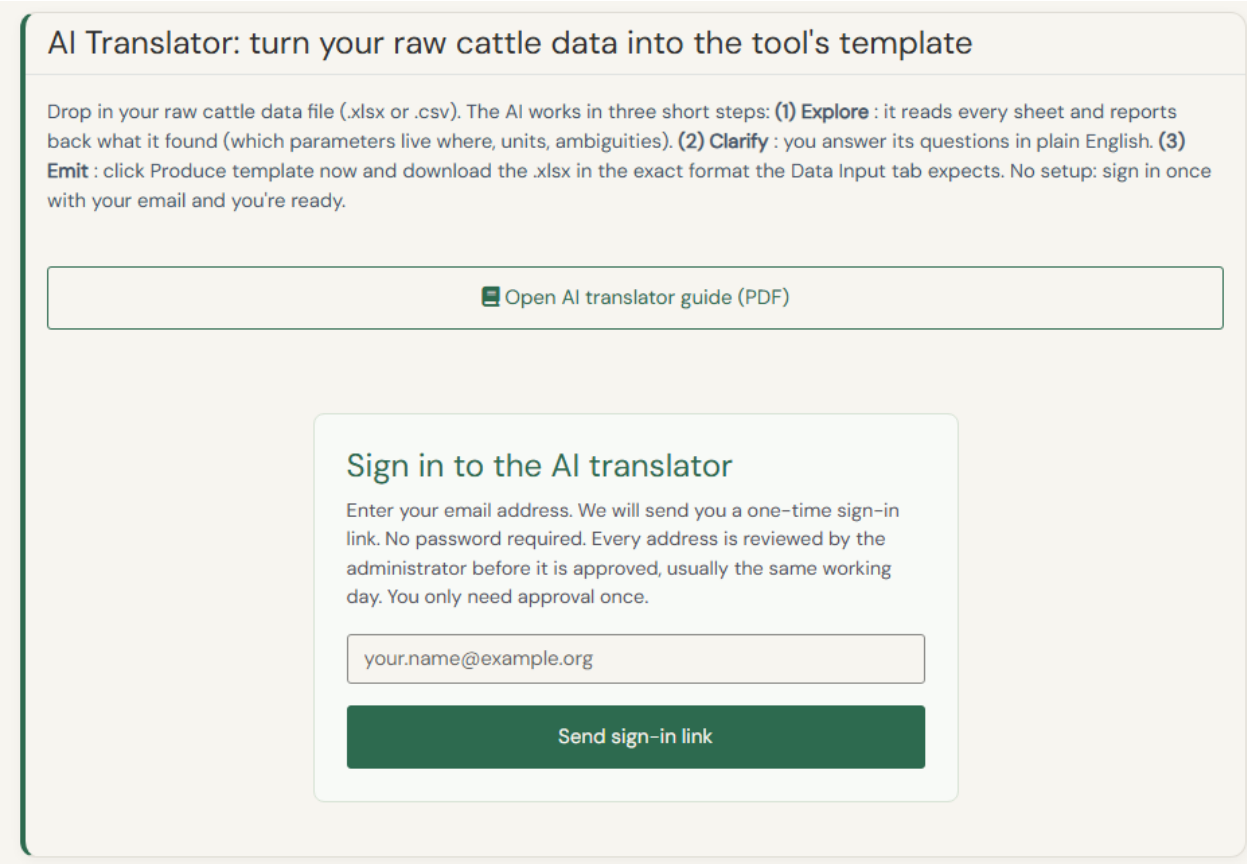
The translator is behind a sign-in by email link. Every address needs a one-time approval by the administrator, including CGIAR addresses, because each translation is paid for by the project. Approval is needed once; after that the browser stays signed in. The full sequence:

15.1 Initial request (first sign-in)

1. Open the **Resources** tab of the app and find the AI Translator card (the first card, with the green left edge).



2. Type your email address in the sign-in box and click **Send sign-in link**.



3. Within one or two minutes, an email arrives from `noreply@cattle-uncertainty.app` with the subject "Sign in to the IPCC Cattle GHG Uncertainty tool". The link stays valid for 30 days and can be clicked more than once. After the first successful click, the browser stays signed in, so the link is only needed once per browser.
4. Click the link. The app opens and sees that your address is not yet approved. It shows this message:

   > Thanks: your request to access the AI translator (your.email@example.org) has been forwarded to the administrator for approval. You'll receive an email once approved.

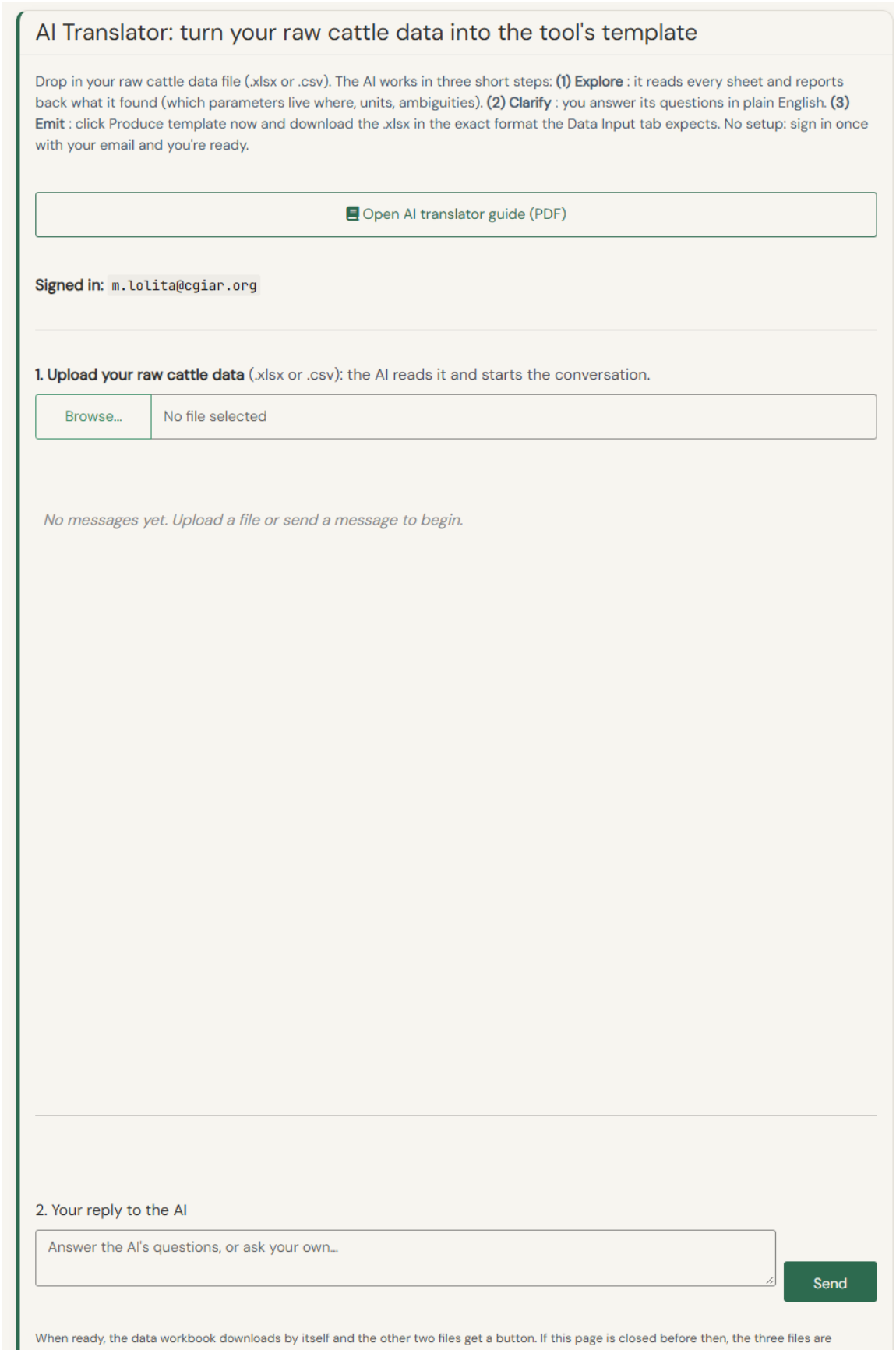
   At the same time, the administrator gets an email with your address and the time of the request.

15.2 Approval (administrator side)

5. The administrator reviews the request, adds the address to the approved list and redeploys the app, usually the same working day.
6. You then get a welcome email with a link back to the app and a short reminder of the workflow.

15.3 Subsequent sign-in (after approval)

7. Go back to the AI Translator card on the Resources tab.
8. Enter the same email address and click **Send sign-in link**. A new link arrives within one or two minutes.
9. Click it. The app checks that the address is approved, sets a long-lived sign-in cookie in the browser, and opens the chat panel.


AI Translator: turn your raw cattle data into the tool's template
Drop in your raw cattle data file (.xlsx or .csv). The AI works in three short steps: (1) Explore : it reads every sheet and reports back what it found (which parameters live where, units, ambiguities). (2) Clarify : you answer its questions in plain English. (3) Emit : click Produce template now and download the .xlsx in the exact format the Data Input tab expects. No setup: sign in once with your email and you're ready.
Open AI translator guide (PDF)
Signed in: m.lolita@cgiar.org
1. Upload your raw cattle data (.xlsx or .csv): the AI reads it and starts the conversation.
Browse...
No file selected
No messages yet. Upload a file or send a message to begin.
2. Your reply to the AI
Answer the AI's questions, or ask your own...
Send
When ready, the data workbook downloads by itself and the other two files get a button. If this page is closed before then, the three files are

10. On later visits from the same browser, the cookie keeps you signed in. No new link is needed unless you clear your site data or change browser.

15.4 Staying signed in

The app remembers you in the browser you signed in with. Another browser or another computer, or clearing your site data, means asking for a new link. A private or incognito window forgets everything when it closes, so it needs a new link every time. Use a normal window, especially if sign-in emails are slow to reach you.

Your conversation is safe either way. It is saved against your email address, not in the browser, so it comes back when you sign in again.

16 When to use the translator and when to skip it

The translator is for the common case where raw inventory data does not match the template. It is most useful when:

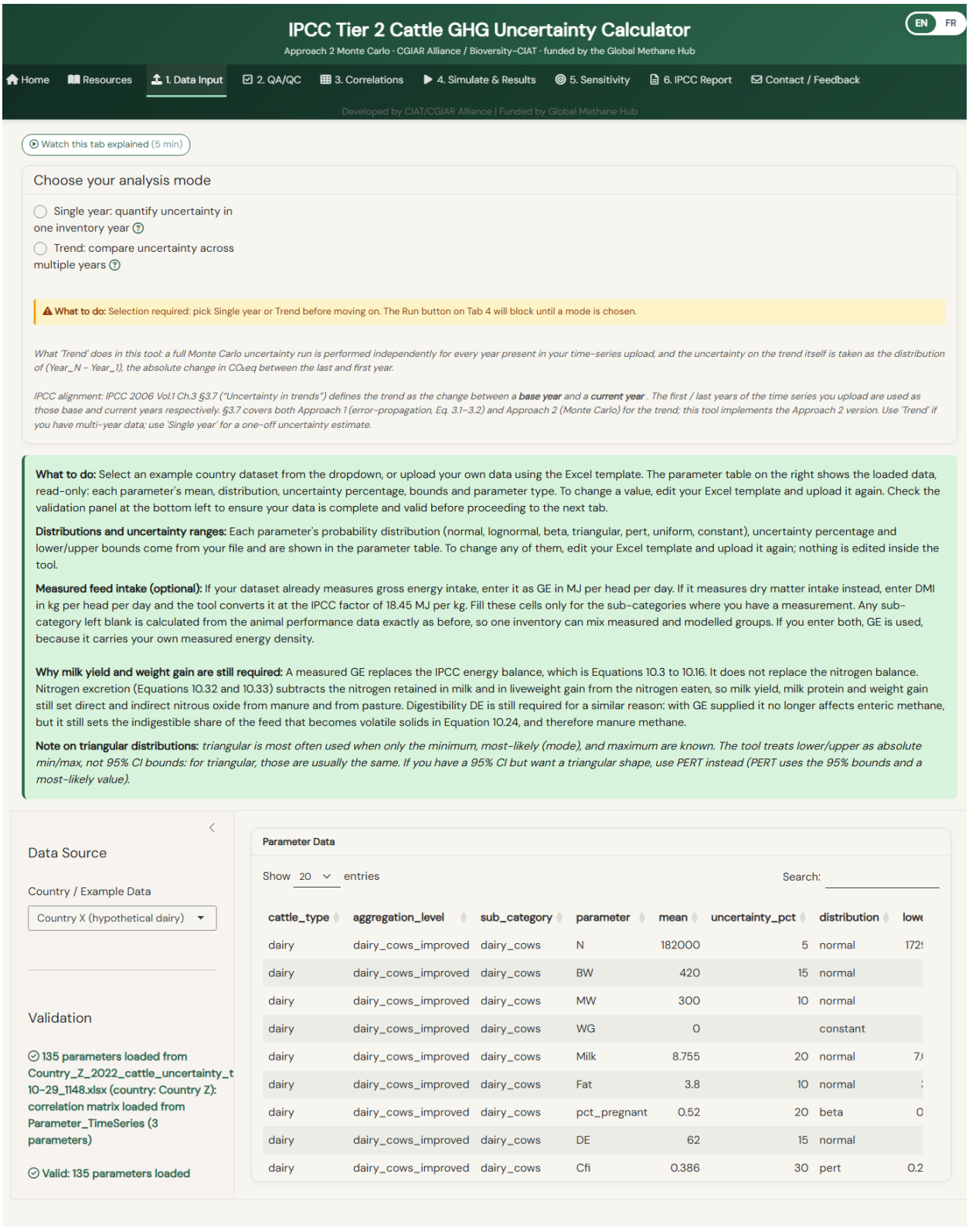
- your data is in an unusual shape (a workbook with section headers, a form-style layout, comments between the values);
- column names differ from the template (`cattle_pop` instead of `N`, `live_wt_lb` instead of `BW`);
- units are mixed (kilograms and pounds, fractions and percentages, litres and kilograms of milk);
- IPCC defaults are needed for parameters you do not measure.

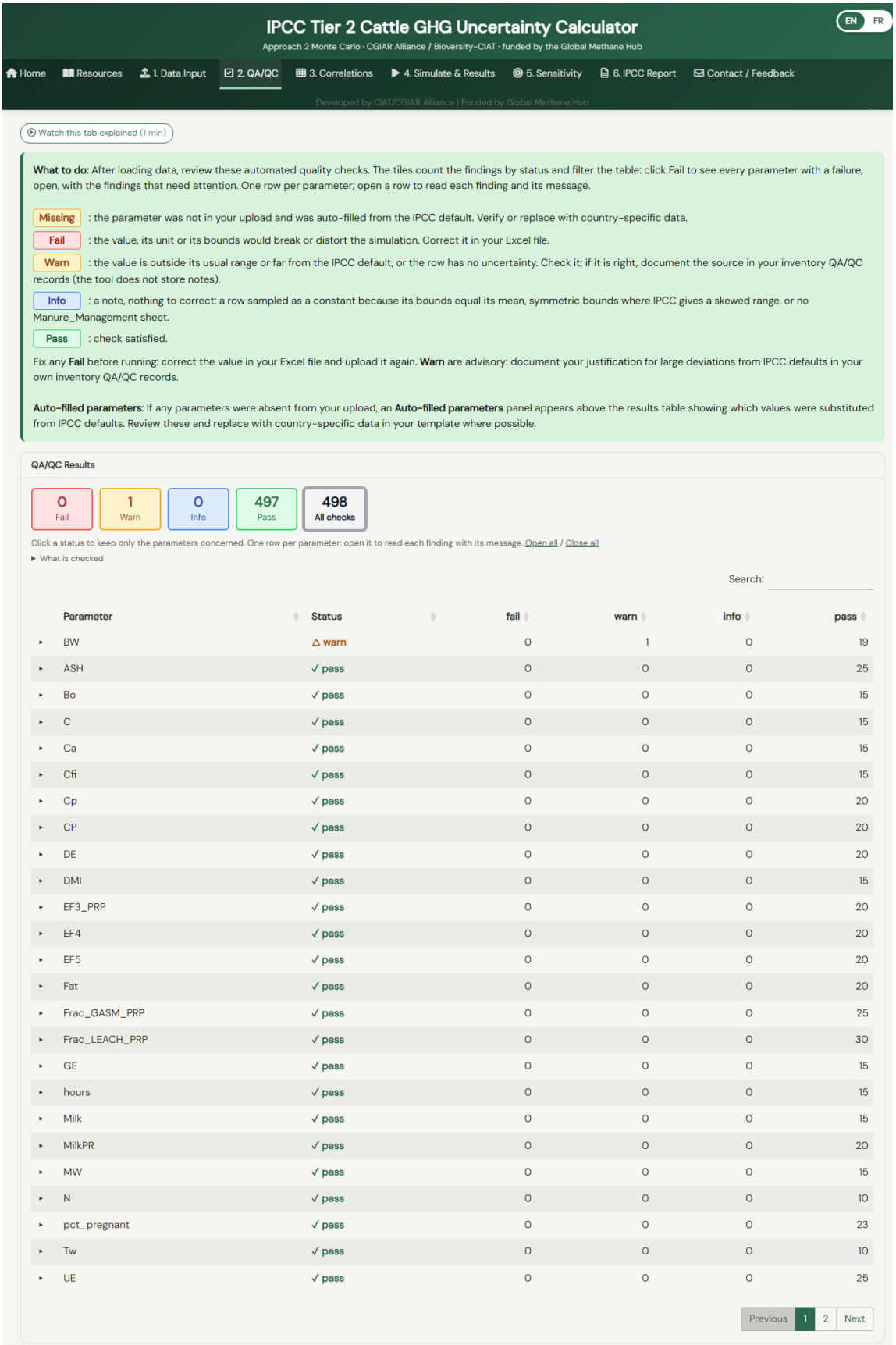
You do not need the translator if you already have a filled template: upload it directly on the Data Input tab. The translator also does not convert emission results; it works on input parameters only.

17 Quality control and limitations

The translator maps columns and converts units. It does not check your values against outside reference data, and it cannot find mistakes in the source file itself (a typing error in the original survey, for example). After every translation:

- **Read the quality preview before downloading.** It lists the checks that will fail after upload and the consistency notes described under Step 3. On the first real translation (a 27-group national inventory), every error the AI had introduced appeared in that list.
- **Spot-check the key numbers.** Compare population, body weight and milk yield in the downloaded template with the corresponding cells in the source file. Any difference is a translation error to raise in the chat.
- **Upload the template on the Data Input tab and review the QA/QC tab.** Every value the translator filled from an IPCC default is marked `ipcc_default` there, so you can decide whether the default suits your country. For body weight (`BW`), the QA tab also compares the value with the continental IPCC default (Annex Tables 10A.1 to 10A.3) and warns on differences above 50%.


IPCC Tier 2 Cattle GHG Uncertainty Calculator
Approach 2 Monte Carlo · CGIAR Alliance / Bioversity-CIAT · funded by the Global Methane Hub
EN FR
Home
Resources
1. Data Input
2. QA/QC
3. Correlations
4. Simulate & Results
5. Sensitivity
6. IPCC Report
Contact / Feedback
Developed by CIAT/CGIAR Alliance | Funded by Global Methane Hub
Watch this tab explained (5 min)
Choose your analysis mode
Single year: quantify uncertainty in one inventory year
Trend: compare uncertainty across multiple years
What to do: Selection required: pick Single year or Trend before moving on. The Run button on Tab 4 will block until a mode is chosen.
What 'Trend' does in this tool: a full Monte Carlo uncertainty run is performed independently for every year present in your time-series upload, and the uncertainty on the trend itself is taken as the distribution of (Year_N - Year_1), the absolute change in CO2eq between the last and first year.
IPCC alignment: IPCC 2006 Vol.1 Ch.3 §3.7 ("Uncertainty in trends") defines the trend as the change between a base year and a current year . The first / last years of the time series you upload are used as those base and current years respectively. §3.7 covers both Approach 1 (error-propagation, Eq. 3.1–3.2) and Approach 2 (Monte Carlo) for the trend; this tool implements the Approach 2 version. Use 'Trend' if you have multi-year data; use 'Single year' for a one-off uncertainty estimate.
What to do: Select an example country dataset from the dropdown, or upload your own data using the Excel template. The parameter table on the right shows the loaded data, read-only: each parameter's mean, distribution, uncertainty percentage, bounds and parameter type. To change a value, edit your Excel template and upload it again. Check the validation panel at the bottom left to ensure your data is complete and valid before proceeding to the next tab.
Distributions and uncertainty ranges: Each parameter's probability distribution (normal, lognormal, beta, triangular, pert, uniform, constant), uncertainty percentage and lower/upper bounds come from your file and are shown in the parameter table. To change any of them, edit your Excel template and upload it again; nothing is edited inside the tool.
Measured feed intake (optional): If your dataset already measures gross energy intake, enter it as GE in MJ per head per day. If it measures dry matter intake instead, enter DMI in kg per head per day and the tool converts it at the IPCC factor of 18.45 MJ per kg. Fill these cells only for the sub-categories where you have a measurement. Any sub-category left blank is calculated from the animal performance data exactly as before, so one inventory can mix measured and modelled groups. If you enter both, GE is used, because it carries your own measured energy density.
Why milk yield and weight gain are still required: A measured GE replaces the IPCC energy balance, which is Equations 10.3 to 10.16. It does not replace the nitrogen balance. Nitrogen excretion (Equations 10.32 and 10.33) subtracts the nitrogen retained in milk and in liveweight gain from the nitrogen eaten, so milk yield, milk protein and weight gain still set direct and indirect nitrous oxide from manure and from pasture. Digestibility DE is still required for a similar reason: with GE supplied it no longer affects enteric methane, but it still sets the indigestible share of the feed that becomes volatile solids in Equation 10.24, and therefore manure methane.
Note on triangular distributions: triangular is most often used when only the minimum, most-likely (mode), and maximum are known. The tool treats lower/upper as absolute min/max, not 95% CI bounds: for triangular, those are usually the same. If you have a 95% CI but want a triangular shape, use PERT instead (PERT uses the 95% bounds and a most-likely value).
Data Source
Country / Example Data
Country X (hypothetical dairy)
Validation
135 parameters loaded from Country_Z_2022_cattle_uncertainty_t 10-29_1148.xlsx (country: Country Z): correlation matrix loaded from Parameter_TimeSeries (3 parameters)
Valid: 135 parameters loaded
Parameter Data
Show 20 entries
Search:
cattle_type aggregation_level sub_category parameter mean uncertainty_pct distribution lowe
dairy dairy_cows_improved dairy_cows N 182000 5 normal 172
dairy dairy_cows_improved dairy_cows BW 420 15 normal
dairy dairy_cows_improved dairy_cows MW 300 10 normal
dairy dairy_cows_improved dairy_cows WG 0 constant
dairy dairy_cows_improved dairy_cows Milk 8.755 20 normal 7.
dairy dairy_cows_improved dairy_cows Fat 3.8 10 normal
dairy dairy_cows_improved dairy_cows pct_pregnant 0.52 20 beta 0
dairy dairy_cows_improved dairy_cows DE 62 15 normal
dairy dairy_cows_improved dairy_cows Cfi 0.386 30 pert 0.2



IPCC Tier 2 Cattle GHG Uncertainty Calculator

Approach 2 Monte Carlo · CGIAR Alliance / Bioversity-CIAT · funded by the Global Methane Hub

EN FR

Home | Resources | 1. Data Input | 2. QA/QC | 3. Correlations | 4. Simulate & Results | 5. Sensitivity | 6. IPCC Report | Contact / Feedback

Developed by CIAT/CGIAR Alliance | Funded by Global Methane Hub

Watch this tab explained (1 min)

What to do: After loading data, review these automated quality checks. The tiles count the findings by status and filter the table: click Fail to see every parameter with a failure, open, with the findings that need attention. One row per parameter; open a row to read each finding and its message.

Missing : the parameter was not in your upload and was auto-filled from the IPCC default. Verify or replace with country-specific data.

Fail : the value, its unit or its bounds would break or distort the simulation. Correct it in your Excel file.

Warn : the value is outside its usual range or far from the IPCC default, or the row has no uncertainty. Check it; if it is right, document the source in your inventory QA/QC records (the tool does not store notes).

Info : a note, nothing to correct: a row sampled as a constant because its bounds equal its mean, symmetric bounds where IPCC gives a skewed range, or no Manure_Management sheet.

Pass : check satisfied.

Fix any **Fail** before running: correct the value in your Excel file and upload it again. **Warn** are advisory: document your justification for large deviations from IPCC defaults in your own inventory QA/QC records.

Auto-filled parameters: If any parameters were absent from your upload, an **Auto-filled parameters** panel appears above the results table showing which values were substituted from IPCC defaults. Review these and replace with country-specific data in your template where possible.

QA/QC Results

0 Fail | 1 Warn | 0 Info | 497 Pass | 498 All checks

Click a status to keep only the parameters concerned. One row per parameter: open it to read each finding with its message. Open all / Close all

▶ What is checked

Search:

Parameter	Status	fail	warn	info	pass
BW	△ warn	0	1	0	19
ASH	✓ pass	0	0	0	25
Bo	✓ pass	0	0	0	15
C	✓ pass	0	0	0	15
Ca	✓ pass	0	0	0	15
Cfi	✓ pass	0	0	0	15
Cp	✓ pass	0	0	0	20
CP	✓ pass	0	0	0	20
DE	✓ pass	0	0	0	20
DMI	✓ pass	0	0	0	15
EF3_PRP	✓ pass	0	0	0	20
EF4	✓ pass	0	0	0	20
EF5	✓ pass	0	0	0	20
Fat	✓ pass	0	0	0	20
Frac_GASM_PRP	✓ pass	0	0	0	25
Frac_LEACH_PRP	✓ pass	0	0	0	30
GE	✓ pass	0	0	0	15
hours	✓ pass	0	0	0	15
Milk	✓ pass	0	0	0	15
MilkPR	✓ pass	0	0	0	20
MW	✓ pass	0	0	0	15
N	✓ pass	0	0	0	10
pct_pregnant	✓ pass	0	0	0	23
Tw	✓ pass	0	0	0	10
UE	✓ pass	0	0	0	25

Previous 1 2 Next

- **The workbook records where it came from.** The Notes cell of `Inventory_Metadata` carries the app version, a fingerprint of the instructions the AI worked from, a fingerprint of the IPCC defaults table, the model and the date, so a file that comes back for support can be traced to the defaults it was built with.
- **Resumed conversations are checked against the current defaults.** If the IPCC defaults or the template changed since a saved conversation, the panel says so and asks for the file to be uploaded again before a template can be produced. Otherwise the earlier discussion would carry the old numbers into the workbook.

18 Privacy and operational cost

18.1 Where your data goes

When you upload a file, its contents are sent to the AI service over an encrypted connection. That is how the translation is done, and it is the main thing to know before uploading anything.

A copy of the conversation, with the uploaded data and your email address, is then kept in a file on the app server, so that you can close the tab and come back later. It is removed in two cases: when you click **Reset conversation**, and when the server restarts after a period without use. There is no scheduled deletion beyond that, so **Reset conversation is the reliable way to remove it**.

Separately, a usage log records the date, your email address, the model and the number of tokens of each call. It holds no file contents and exists so the administrator can watch the spending.

The AI service applies its own terms to the data it receives. Those are the provider's terms and should be read directly rather than summarised here.

18.2 Cost

The translator runs on an AI account paid for by the GMH project. You are not billed. A monthly spending cap applies; if it is reached, a "monthly cap reached, contact administrator" message appears and translations pause until the next month.

18.3 Whether to use it at all

The inventory data the translator handles is generally not sensitive, because it is meant for UNFCCC reporting, which is public. If your ministry has a policy against sending pre-publication data to an external AI service, do not use the translator. Fill the template by hand instead, with the IPCC defaults given in the user guide.

18.4 Related material

- User Guide, section "AI-assisted data preparation": the same workflow in the printable user guide, including the step-by-step for non-CGIAR addresses.
- Methodology document: the equations, the parameter definitions and the references for everything the translator writes.
- Find out more: Correlations: the companion page on the correlation modes of the Simulate tab.